# Interfacing Input/Output Devices with FPGA

**Objective**

The main objective of this experiment is to understand how different input and output (I/O) devices can be connected and controlled using the **ALTERA Cyclone II EP2C5T144C8 FPGA**. In this experiment, students learn how to work with external devices such as switches, LEDs, and seven-segment displays through the FPGA's GPIO pins. The experiment also focuses on assigning and configuring I/O pins, handling digital signals, and connecting external devices to the FPGA logic.

**2. Introduction to FPGA I/O Interfacing**

**2.1. I/O Standards and Configuration**

FPGAs provide different I/O standards so that they can communicate properly with various external electronic devices. The **Cyclone II EP2C5T144C8** supports standards such as LVTTL, LVCMOS, SSTL, and HSTL. These standards define the voltage levels and electrical characteristics used by the FPGA's I/O pins.

- **LVTTL (Low Voltage TTL):** This standard is commonly used with TTL-based digital circuits and typically operates at a 3.3V logic level.
- **LVCMOS (Low Voltage CMOS):** LVCMOS supports different voltage levels, including 1.8V, 2.5V, and 3.3V. It is widely used for connecting FPGA devices with modern digital circuits.
- **SSTL (Stub Series Terminated Logic) and HSTL (High-Speed Transceiver Logic):** These standards are mainly used in high-speed applications, particularly memory interfaces and other systems that require reliable high-speed data transfer.

**2.2. Configuring I/O Pins**

Each I/O pin on the Cyclone II FPGA can be configured according to the requirements of the design. A pin can operate as an input, output, or bidirectional connection. These settings, along with the required pin assignments and I/O standards, can be configured using **Quartus II**.

- **Input Mode:** In this mode, the FPGA receives digital signals from an external device such as a switch or sensor.
- **Output Mode:** In output mode, the FPGA sends digital signals to an external device such as an LED or display.
- **Bidirectional Mode:** A bidirectional pin can work as either an input or an output, depending on how it

is configured in the FPGA design.

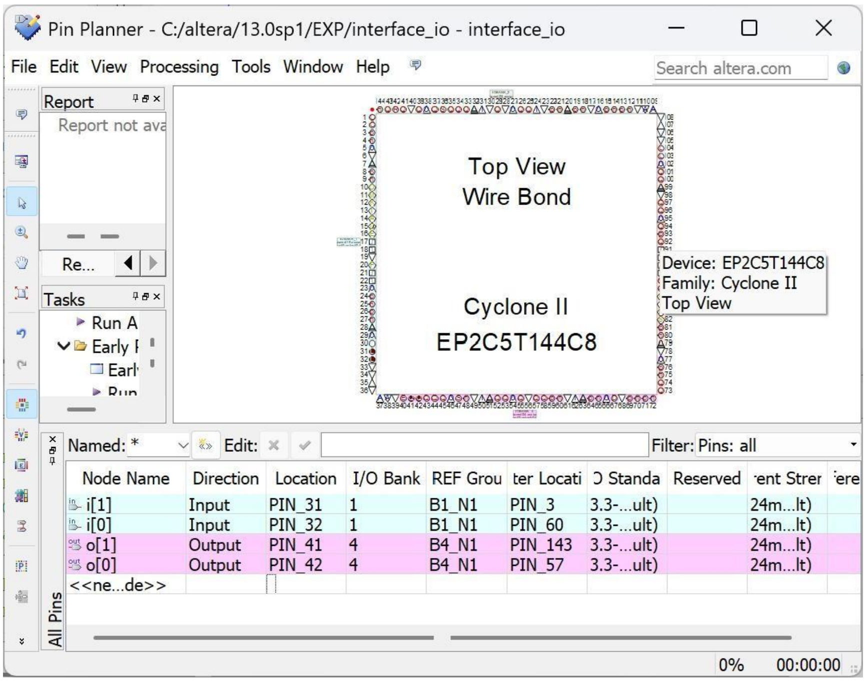
### 3. Interfacing with LEDs

#### 3.1. Overview

LEDs are commonly used with FPGA boards to provide simple visual feedback. They can indicate the current status of a circuit, display the result of a logic operation, or help during debugging. LEDs are connected to GPIO pins that are configured as outputs.

Push buttons, on the other hand, are used to provide digital input signals to the FPGA. They are connected to GPIO pins configured as inputs. When a button is pressed or released, the FPGA detects the corresponding change in the input signal and can perform the required operation.

#### 3.2. Pin Planning




#### 2.1. Technical Details

- **Connection: Each push button is connected to a GPIO input pin of the FPGA. A pull-down resistor can be used to ensure that the input remains at a stable LOW level when the button is not pressed.**
- **Debouncing: Mechanical push buttons can produce unwanted rapid changes in the signal when they are pressed or released. A debouncing technique can be added to the design to remove these unwanted transitions and provide a stable input signal.**

#### 3.4. Example Design

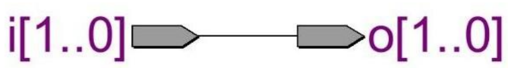
**Push Button to LED Control – Pin Assignments in Quartus II**

**The design connects the push buttons to FPGA input pins and the LEDs to output pins. The FPGA processes the button signals and controls the corresponding LEDs based on the programmed logic.**

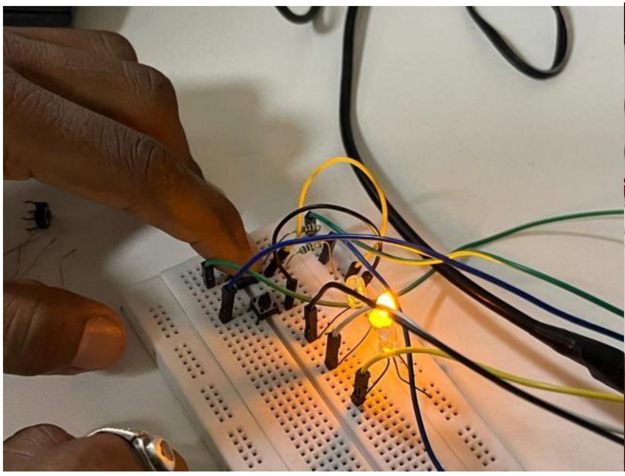
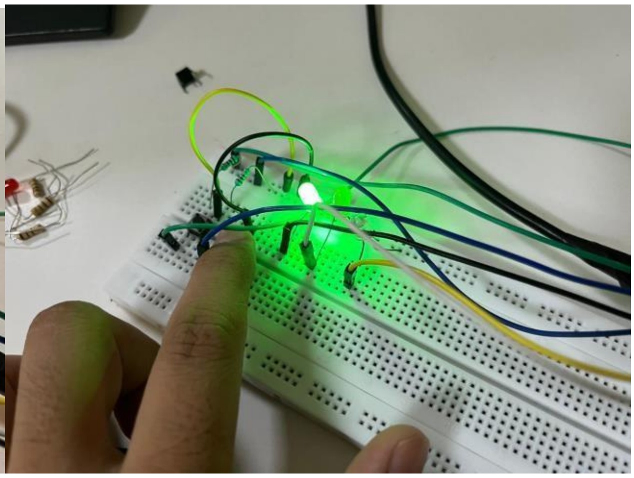
**3.5. Implementation and Testing**

- **Synthesis and Implementation: The design is first compiled in Quartus II. During compilation, the software checks the design and implements it using the available FPGA resources. Correct pin assignments should be verified before programming the board.**
- **Programming: Once the design has been successfully compiled, the generated programming file is downloaded to the FPGA using the Quartus II Programmer tool.**
- **Testing: After programming, each push button is pressed individually to check whether the corresponding LED responds correctly. This confirms that the input and output connections have been properly configured.**

**3.6 RTL schematic:**




**3.7 Output**

## Conclusion

In this experiment, we learned how to interface input and output devices with the **Altera Cyclone II FPGA** using **Quartus II**. We configured the GPIO pins, assigned the required FPGA pins, compiled the design, and programmed it onto the FPGA board.

Push buttons were used as input devices, while LEDs were used as output devices. The practical results showed that pressing a particular push button could control the corresponding LED as expected. Through this experiment, we gained a better understanding of FPGA I/O interfacing, pin assignment, digital signal handling, and hardware implementation. These concepts are important for developing practical FPGA-based digital systems.

**Post Lab Exercise**

1. What is the primary function of GPIO pins in an FPGA?
   GPIO (General Purpose Input/Output) pins are used to connect the FPGA with external devices. They can be configured as input, output, or bidirectional pins to receive or send digital signals.

2. What is the purpose of input switches on an FPGA development board?
   Input switches provide manual digital input signals to the FPGA. They allow users to test and verify FPGA logic by changing the input values during hardware execution.

3. Name one practical application of FPGA I/O interfacing.
   One practical application is **industrial automation**, where an FPGA interfaces with sensors and actuators to monitor and control machines in real time.
   **Other examples:**
   - Traffic light control systems
   - Robotics
   - Home automation
   - Medical devices

4. What is the purpose of the Programmer tool in Quartus II?
   The Programmer tool in Quartus II is used to download the compiled configuration file (.sof) into the FPGA device. It programs the FPGA so that the implemented digital circuit can operate on the hardware.

5. Why is pin assignment necessary before programming an FPGA?
   Pin assignment ensures that the logical input and output signals in the design are mapped to the correct physical FPGA pins. Without proper pin assignments, the FPGA cannot communicate correctly with external devices such as switches, LEDs, and displays.